

Practice Sheet 1.4: Basics of Python Programming

□ HARD LEVEL — IF-ELSE (Exam-Ready)

1. Income Tax Calculation

Problem Statement:

Write a Python program to calculate the **income tax** based on the following slabs:

- $\text{Income} \leq 2,50,000 \rightarrow \text{No tax}$
- $\text{Income} \leq 5,00,000 \rightarrow 5\% \text{ of income}$
- $\text{Income} \leq 10,00,000 \rightarrow 20\% \text{ of income}$
- $\text{Income} > 10,00,000 \rightarrow 30\% \text{ of income}$

Input Format:

Annual income (integer)

Output Format:

Print the tax amount.

2. Grade with Distinction and Re-Exam

Problem Statement:

Write a Python program to display the **final result** of a student based on marks:

- $\text{Marks} \geq 85 \rightarrow \text{Distinction}$
- $\text{Marks} \geq 60 \rightarrow \text{First Class}$
- $\text{Marks} \geq 50 \rightarrow \text{Second Class}$
- $\text{Marks} \geq 40 \rightarrow \text{Pass}$
- $\text{Marks} \geq 35 \rightarrow \text{Re-Exam}$
- $\text{Below } 35 \rightarrow \text{Fail}$

Input Format:

Marks (integer)

Output Format:

Print the result.

3. ATM Transaction with Charges

Problem Statement:

Write a Python program to check whether an ATM withdrawal is successful and calculate the **remaining balance**.

Conditions:

- Withdrawal amount must be a multiple of 100
- Withdrawal amount + ₹50 transaction charge must be $\leq$ balance

Input Format:

Account balance (integer)

Withdrawal amount (integer)

Output Format:

Print transaction status and remaining balance.

4. Electricity Bill with Fixed Charges

Problem Statement:

Write a Python program to calculate the electricity bill based on slab rates and a fixed charge of ₹100.

- Units $\leq 100 \rightarrow$ ₹1.5 per unit
- Units $\leq 200 \rightarrow$ ₹2.5 per unit
- Units $> 200 \rightarrow$ ₹4 per unit

Input Format:

Units consumed (integer)

Output Format:

Print total bill amount.

5. Loan Eligibility Check

Problem Statement:

Write a Python program to check whether a customer is **eligible for a loan** based on:

- Age between 21 and 60
- Monthly income $\geq 30,000$
- Credit score ≥ 700

Input Format:

Age (integer)

Monthly income (integer)

Credit score (integer)

Output Format:

Print Loan Approved or Loan Rejected.

6. Mobile Data Plan Selection

Problem Statement:

Write a Python program to suggest a mobile data plan based on usage:

- Usage ≤ 1.5 GB/day $\rightarrow$ Basic Plan
- Usage ≤ 3 GB/day $\rightarrow$ Standard Plan
- Usage ≤ 5 GB/day $\rightarrow$ Premium Plan
- Above 5 GB/day $\rightarrow$ Unlimited Plan

Input Format:

Daily data usage in GB (float)

Output Format:

Print the recommended plan.

7. Validate Date

Problem Statement:

Write a Python program to check whether the given **date is valid**.
Consider:

- Valid day range based on month
- February has 28 days (ignore leap year)

Input Format:

Day (int)
Month (int)
Year (int)

Output Format:

Print Valid Date or Invalid Date.

8. Insurance Premium Calculation

Problem Statement:

Write a Python program to calculate the **insurance premium** based on age and health condition:

- Age < 30 and health = "good" → ₹5,000
- Age < 30 and health = "poor" → ₹8,000
- Age ≥ 30 and health = "good" → ₹7,000
- Age ≥ 30 and health = "poor" → ₹10,000

Input Format:

Age (integer)
Health condition (good / poor)

Output Format:

Print the premium amount.

9. University Admission Eligibility

Problem Statement:

Write a Python program to check admission eligibility:

- Maths ≥ 60
- Physics ≥ 55
- Chemistry ≥ 50
- Total of three subjects ≥ 180

Input Format:

Marks in Maths, Physics, Chemistry

Output Format:

Print Eligible or Not Eligible.

10. Calculate Parking Charges

Problem Statement:

Write a Python program to calculate **parking charges**:

- Up to 2 hours $\rightarrow$ ₹20
- 3–5 hours $\rightarrow$ ₹50
- 6–10 hours $\rightarrow$ ₹100
- More than 10 hours $\rightarrow$ ₹200

Input Format:

Number of hours parked (integer)

Output Format:

Print parking charges.